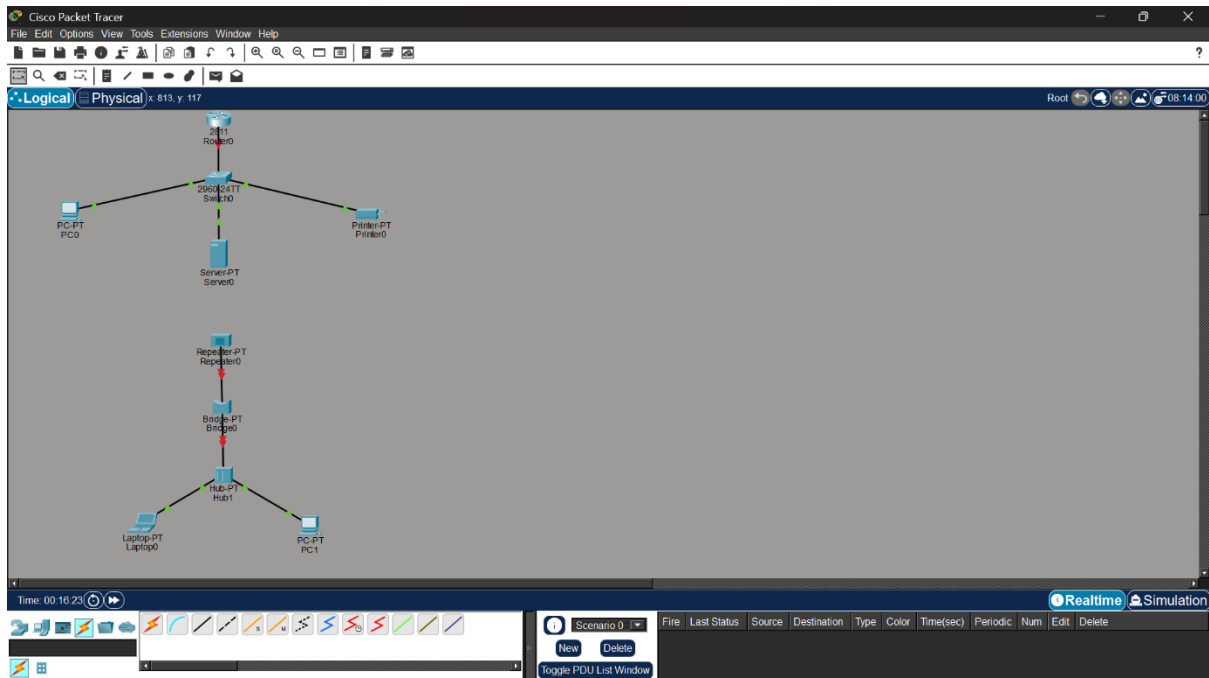
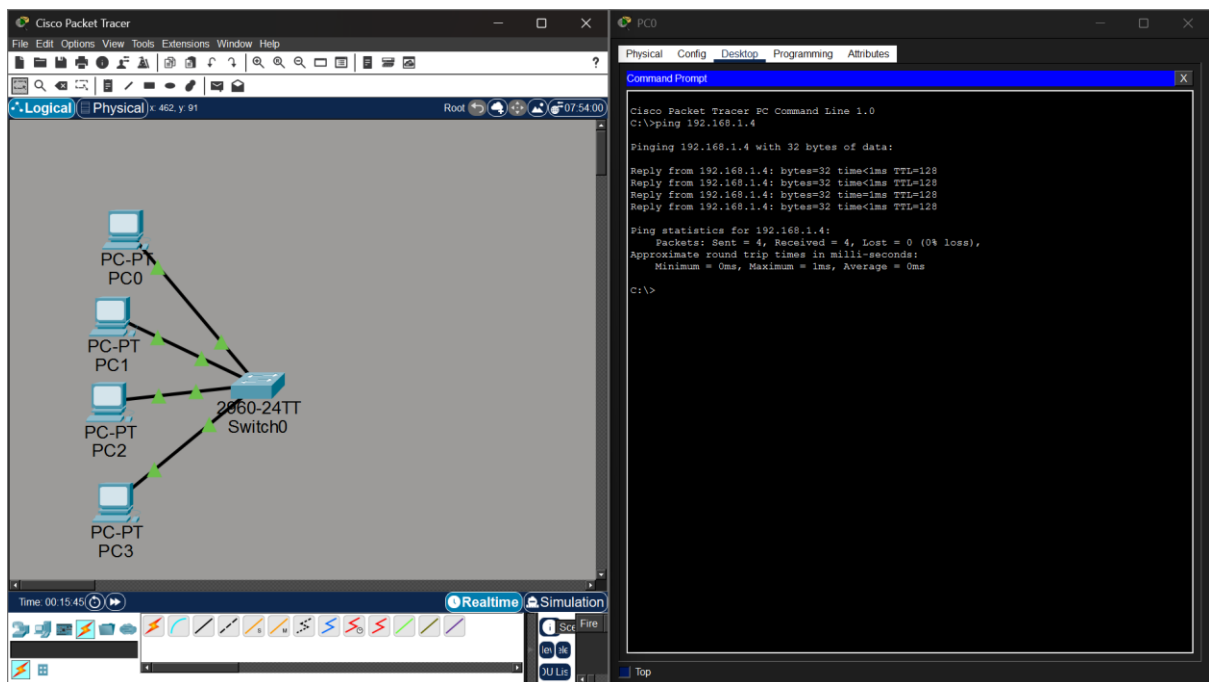


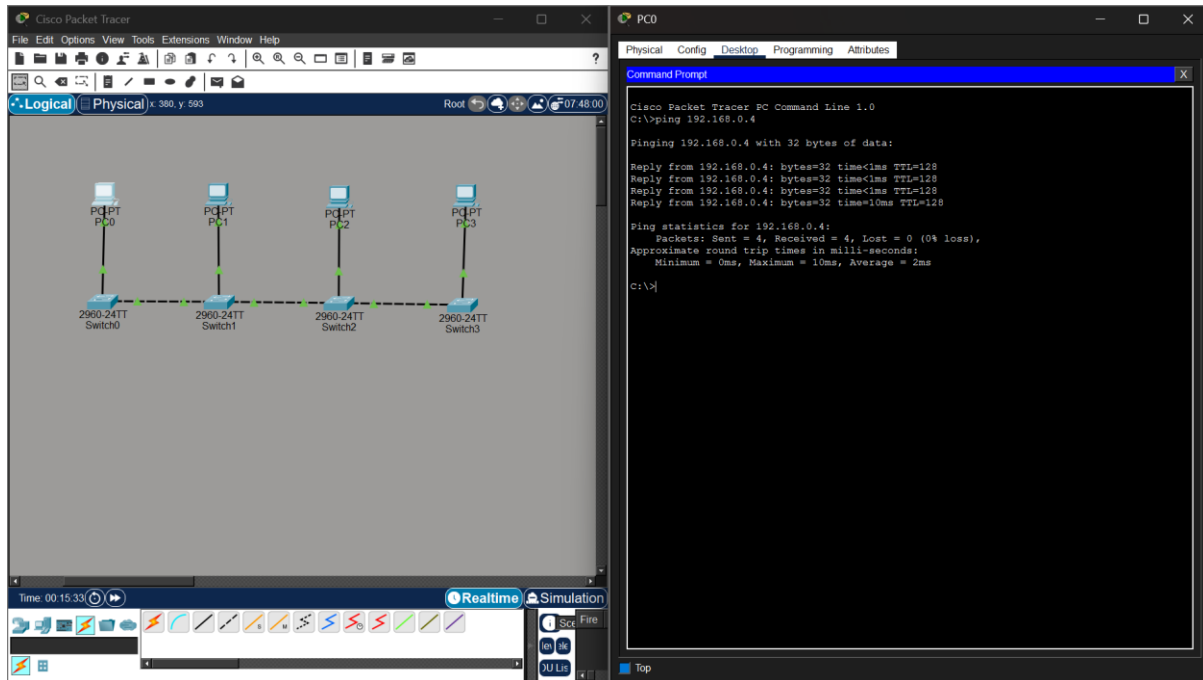
Experiment-01-Network-Components



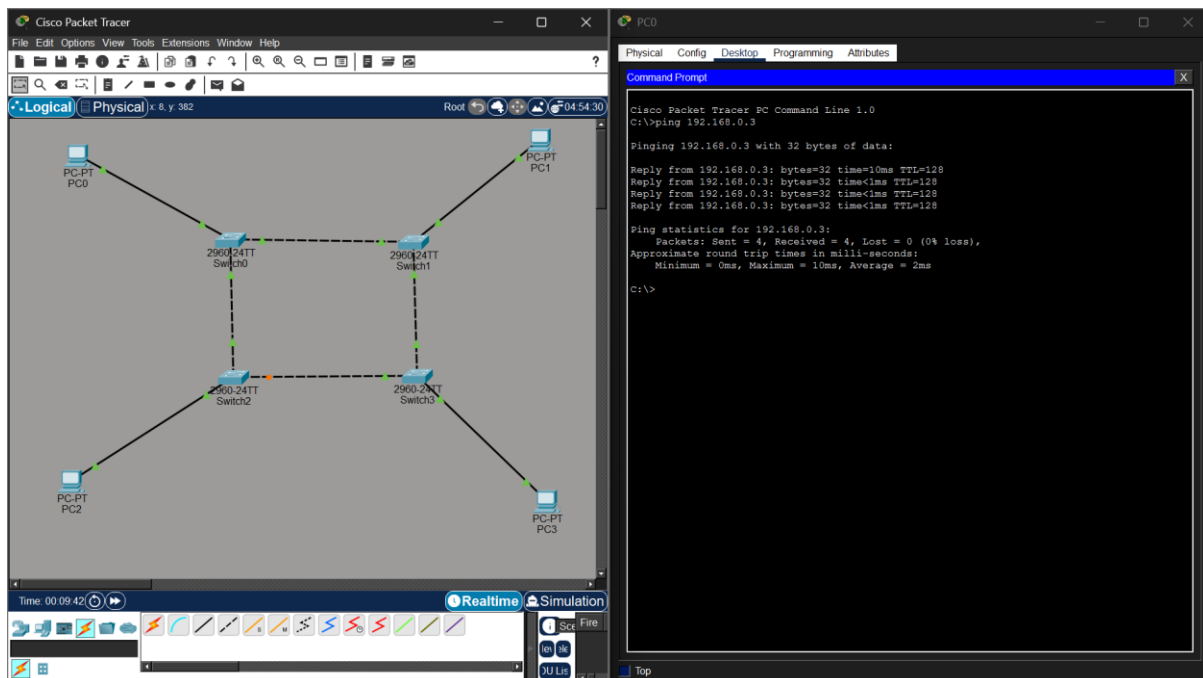
Experiment-02-Star-Topology



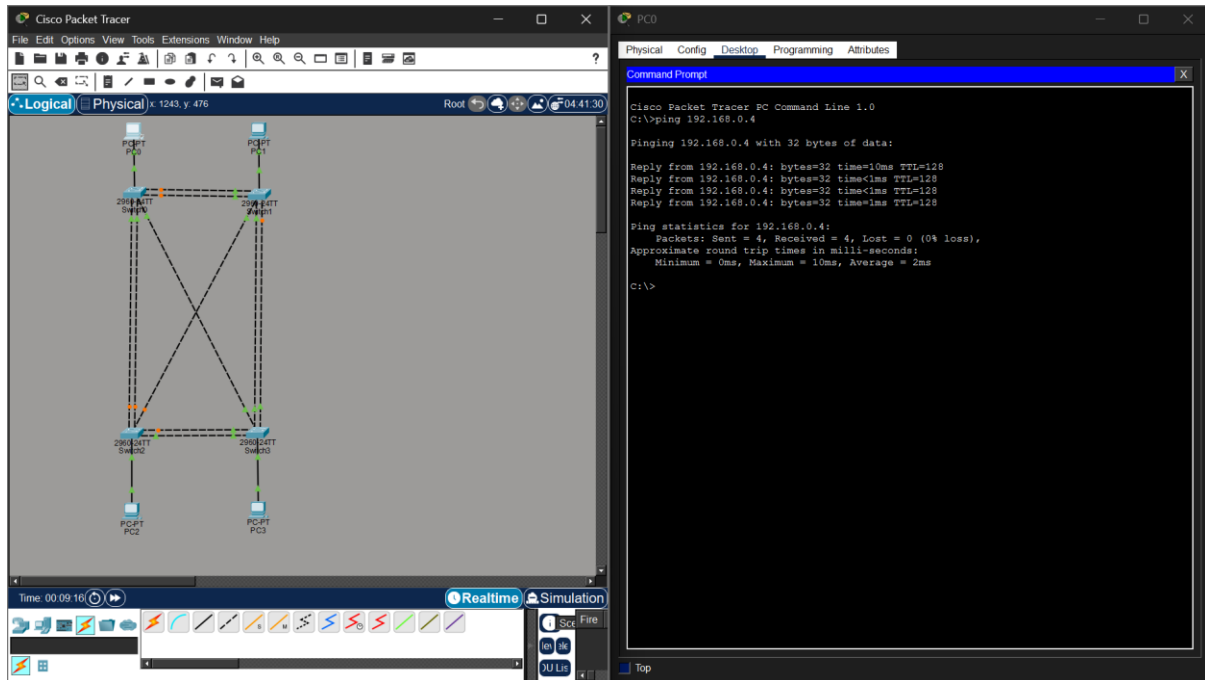
Experiment-03-Bus-Topology



Experiment-04-Ring-Topology



Experiment-05-Mesh-Topology



Experiment-06-Tree-Topology

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1879, y: 983

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
0.005	Switch2	
0.005	PC3	
0.006	Switch2	
0.006	PC4	
0.006	Switch2	
0.007	PC5	
0.007	Switch2	
0.007	Switch2	
0.008	Switch2	
0.008	Switch0	
0.008	Switch1	
0.009	Switch0	
0.009	Switch1	
0.010	Switch1	
0.710	--	

Reset Simulation Constant Delay Capturing

Play Controls

Event List Filters - Visible Events

ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, LACP, LLDP, MDIOBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPv2, RTP, SCP, SNMP, SNMPv3, SSH, SIP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Time: 00:17:34.731 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1539, y: 517

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
0.005	Switch2	
0.005	PC3	
0.006	Switch2	
0.006	PC4	
0.006	Switch2	
0.007	PC5	
0.007	Switch2	
0.007	Switch2	
0.008	Switch2	
0.008	Switch0	
0.008	Switch1	
0.009	Switch0	
0.009	Switch1	
0.010	Switch1	
0.710	--	

Reset Simulation Constant Delay Capturing

Play Controls

Event List Filters - Visible Events

ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, LACP, LLDP, MDIOBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPv2, RTP, SCP, SNMP, SNMPv3, SSH, SIP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

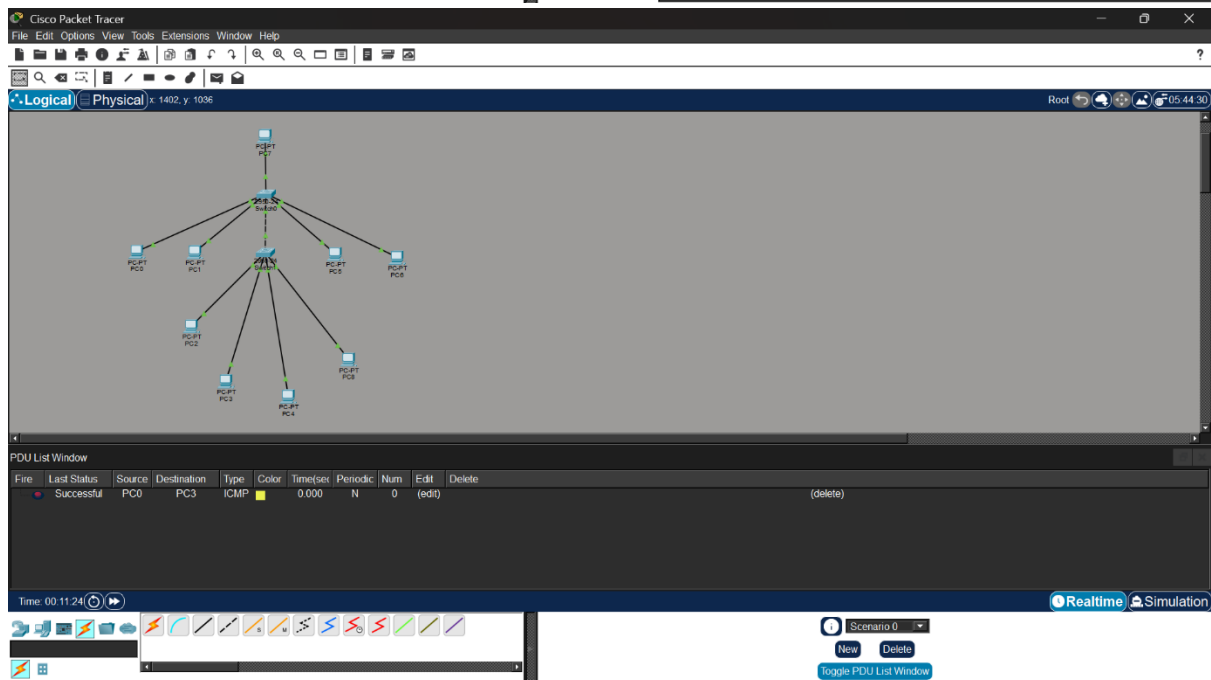
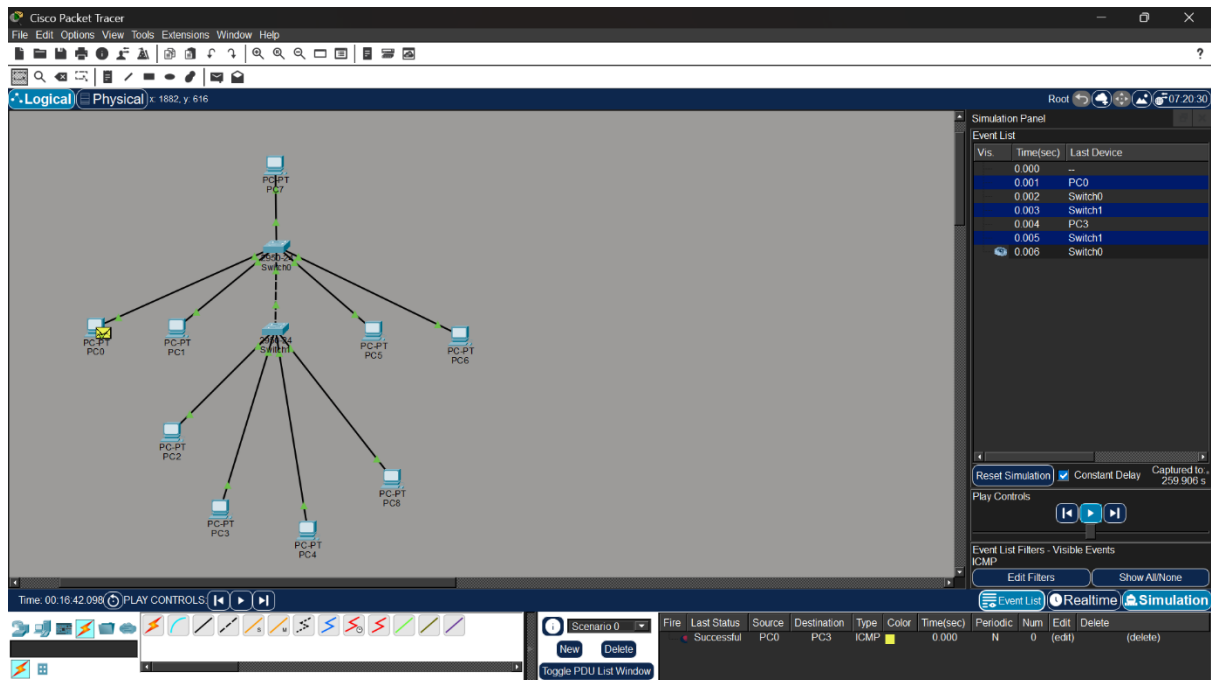
Time: 00:14:57/ PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

Experiment-07-Hybrid-Topology



Experiment-08-ARP-Analysis

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating a network setup and the process of analyzing ARP requests.

Top Screenshot: The network topology shows a central Hub0 connected to six PCs (PC0, PC1, PC2, PC3, PC4, PC5). The PDU List Window is open, showing a successful ICMP packet from PC0 to PC3. The simulation is running in Realtime mode.

Bottom Screenshot: The network topology is the same, but the PDU List Window shows a successful ICMP packet from PC0 to PC3. The Event List Panel is open, showing a list of events. The simulation is running in Realtime mode.

Event List Panel:

Vis.	Time(sec)	Last Device
	0.000	
	0.001	PC0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.003	PC3
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0

PDU List Window:

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)

Experiment-09-CSMA-CD-CSMA-CA

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating a network simulation setup for CSMA-CD and CSMA-CA experiments.

Top Screenshot: The main window shows a network topology with a central switch (SW1) connected to three PCs (PC0, PC1, PC2). The simulation panel on the right shows the event list with timestamps 0.000, 0.001, 0.002, 0.003, and 0.004, corresponding to the devices. The command prompt window on the right shows the configuration of the FastEthernet0 interface, including the link-local IPv6 address (FE80::2D0:FFFF:FED5:9268), IPv6 address (192.168.1.1), IPv4 address (192.168.1.1), subnet mask (255.255.255.0), and default gateway (0.0.0.0). The command prompt also shows the results of a ping command to 192.168.1.2, indicating successful connectivity.

Bottom Screenshot: The main window shows the same network topology. The simulation panel on the right shows the event list with timestamps 0.000, 0.001, 0.002, 0.003, and 0.004, corresponding to the devices. The command prompt window on the right shows the results of a ping command to 192.168.1.3, indicating successful connectivity.

Experiment-10-Computer-Lab

The image displays two screenshots of the Cisco Packet Tracer application. The top screenshot shows a network diagram in the 'Logical' view. A central switch, labeled '2960-24 Switch0', is connected to four PC-PT devices (PC0, PC1, PC2, PC3) and a router labeled '1841 Router0'. The bottom screenshot shows the same network diagram, but with a 'Command Prompt' window open on PC0. The command prompt displays the output of the 'ipconfig' command, showing IPv6 configuration details for the FastEthernet0 interface. It also shows the output of 'ping 192.168.1.254' and 'ping 192.168.1.4' commands, indicating successful connectivity with 0% loss.

Top Screenshot: Network Diagram

- Router: 1841 Router0
- Switch: 2960-24 Switch0
- PCs: PC-PT PC0, PC-PT PC1, PC-PT PC2, PC-PT PC3

Bottom Screenshot: Command Prompt Output

```
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FETA:2C39
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time=12ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

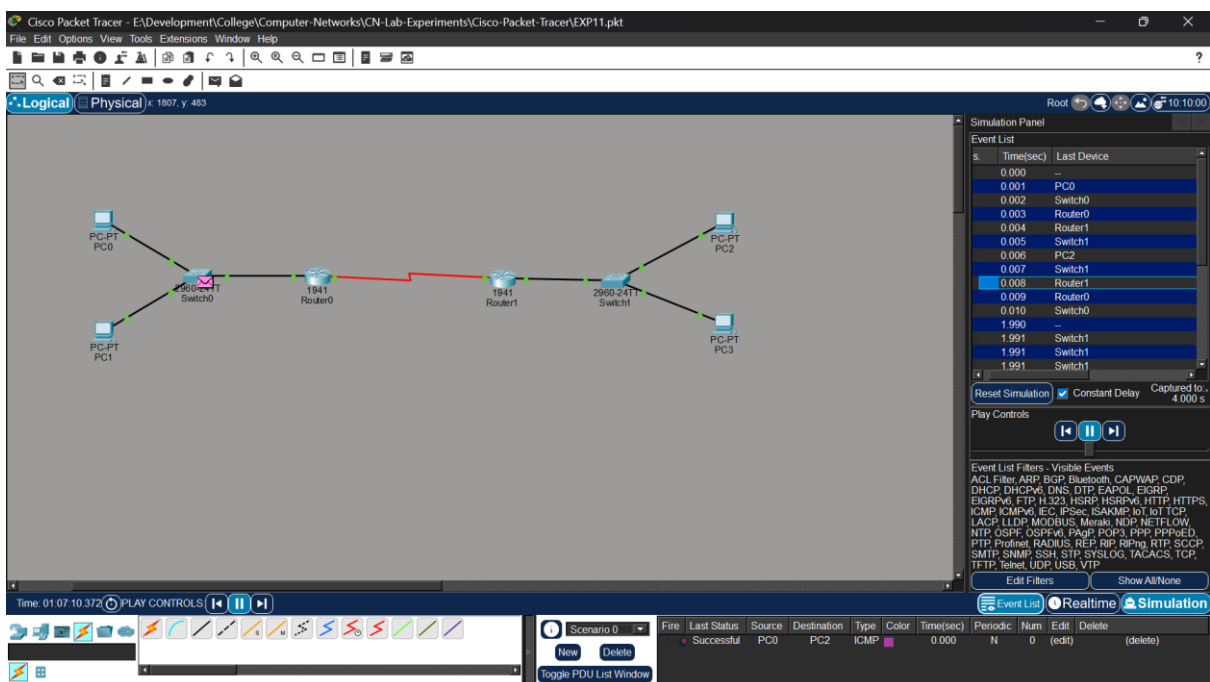
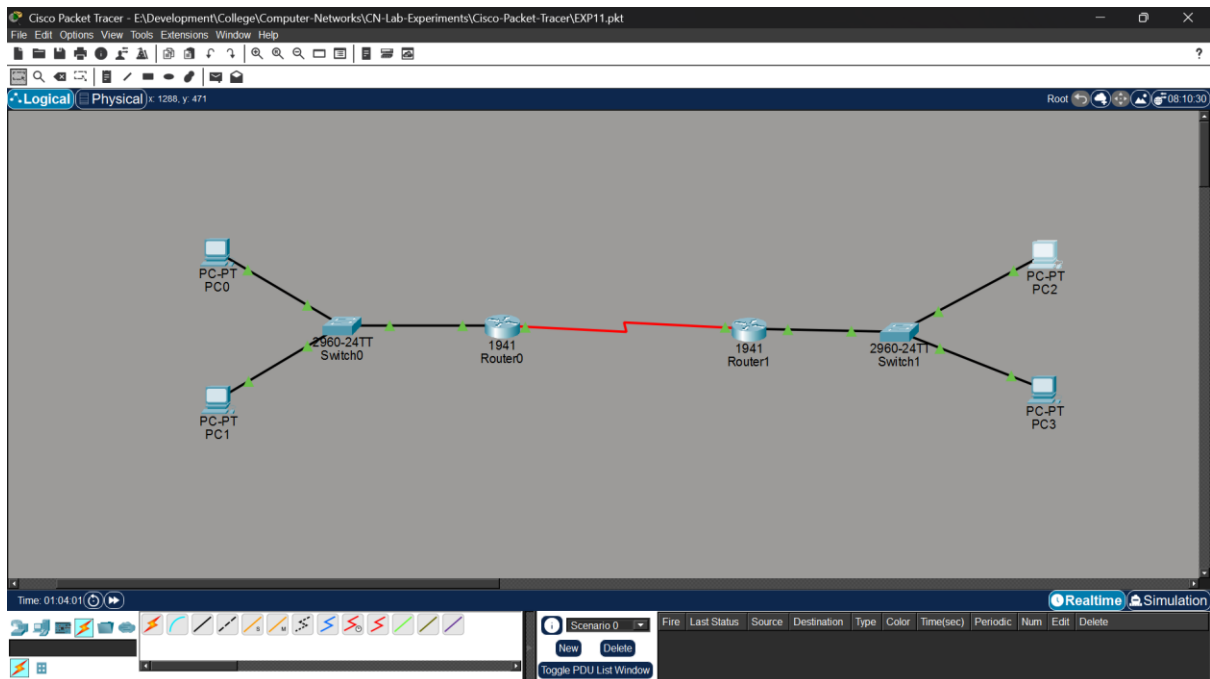
Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

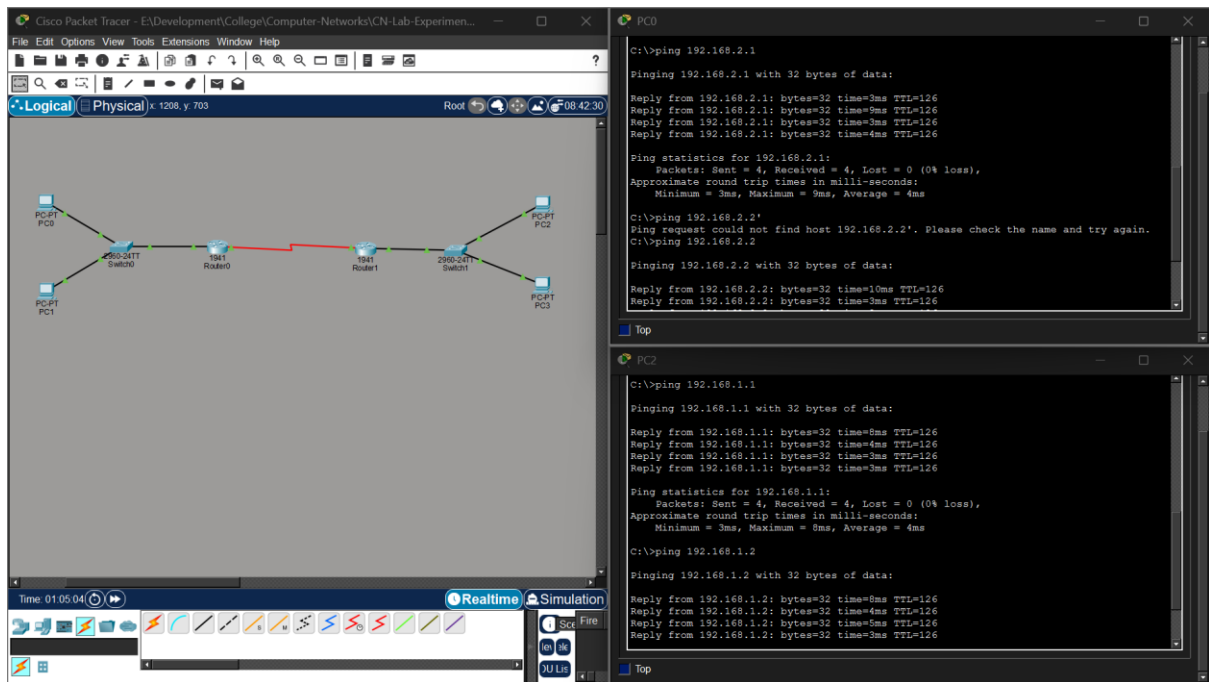
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

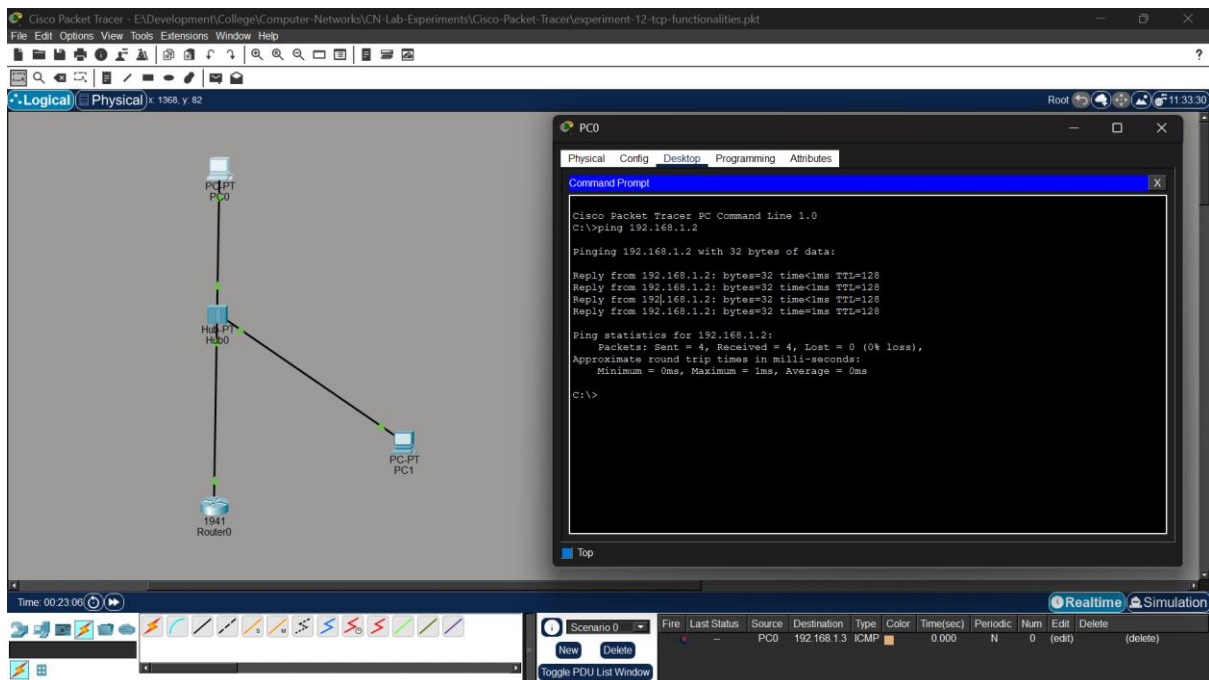
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
```


Experiment-11-Computer-Lab Router to Router Communication Using Static Routing

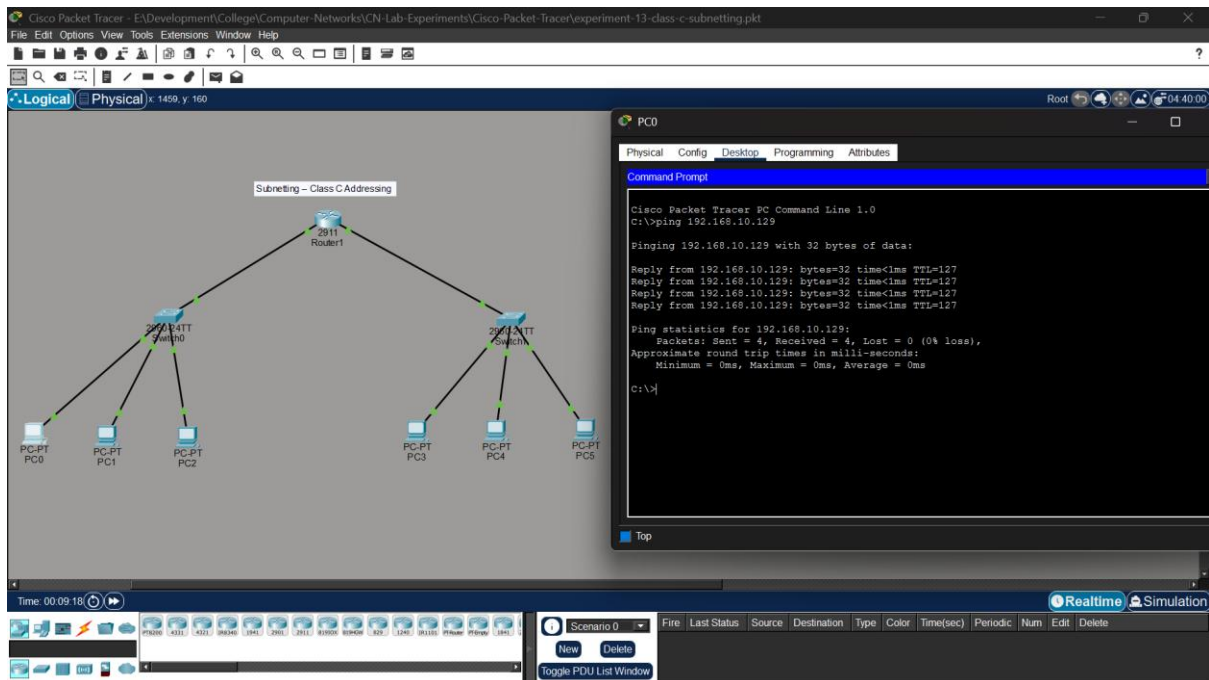




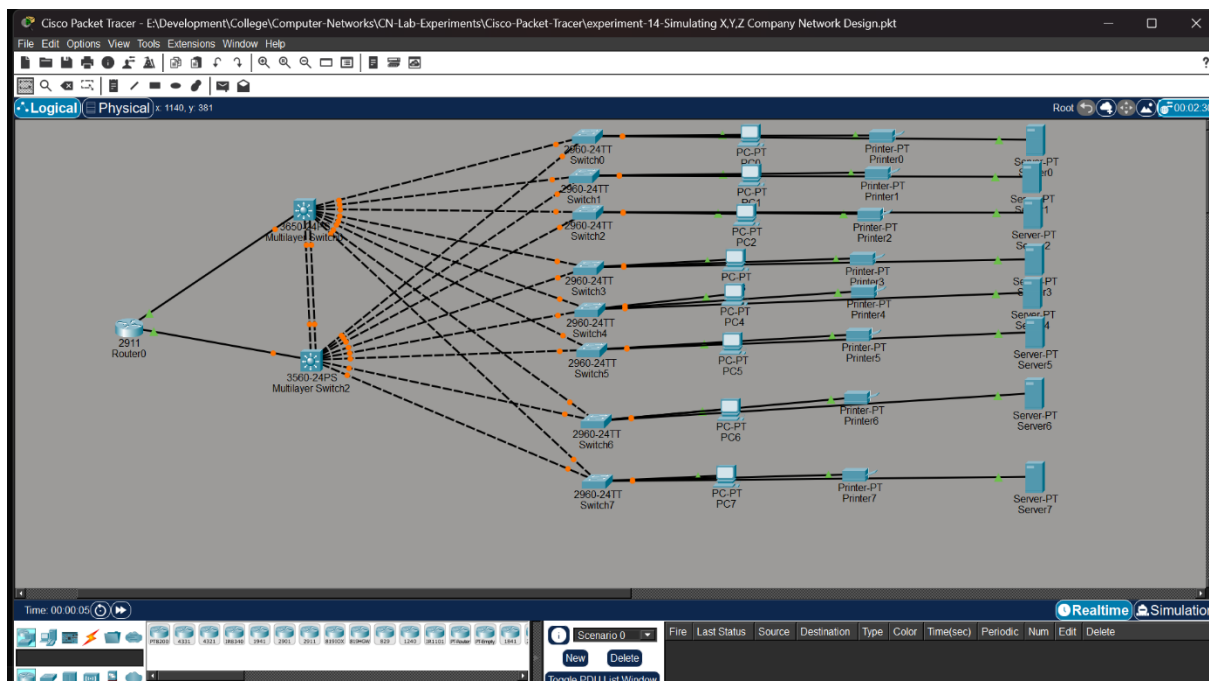
Experiment-12-Tcp-Functionalities



Experiment-13- Experiment-class-c-subnetting



Experiment-14- xyz-company-network-design



Experiment-15- DHCP Configuration

The image displays the Cisco Packet Tracer interface. On the left, a network topology is shown in the 'Logical' view. It consists of a central '2960 24-T Switch' connected to three 'PC-PT' devices (PC0, PC1, PC2) and a 'Server-PT Server0'. The time is 00:09:06. On the right, the 'PC0' configuration window is open, showing the 'Config' tab. The 'IP Configuration' section is set to 'DHCP' (Static is selected, but DHCP request is successful). The 'IPv6 Configuration' section is set to 'Automatic' (Static is selected). The '802.1X' section is set to 'Use 802.1X Security' (MD5 is selected). The 'Fire' section is set to 'Last Status'.

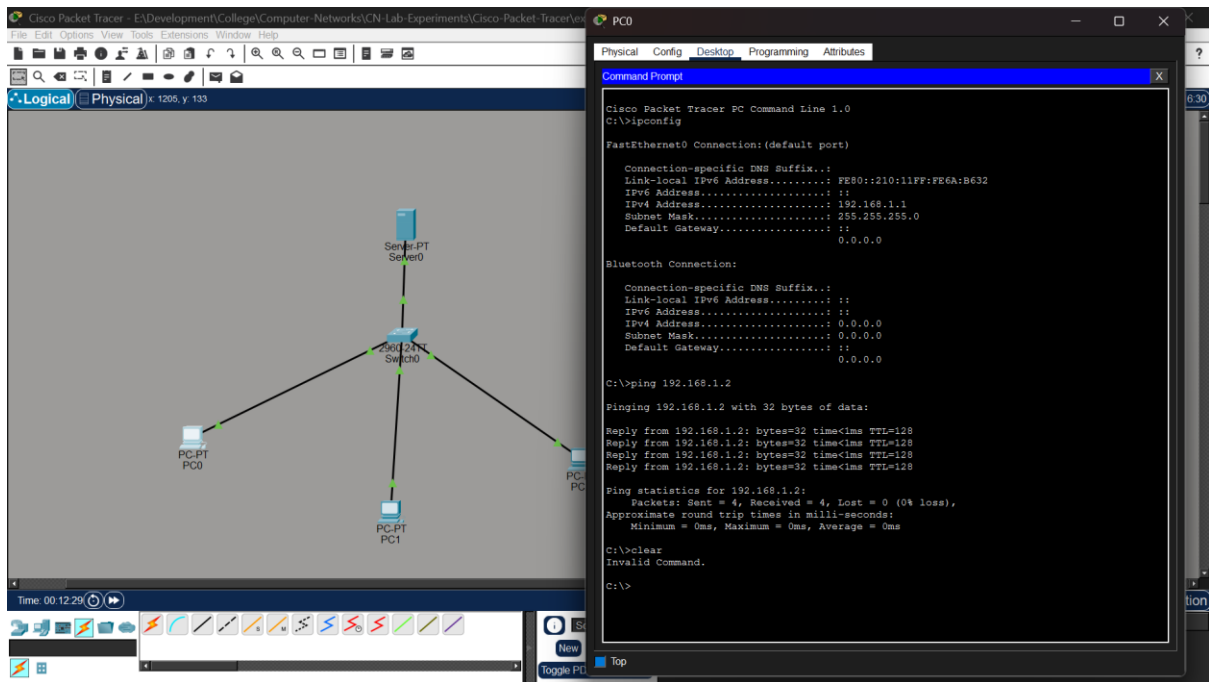
PC0 Configuration:

- Interface: FastEthernet0
- IP Configuration:
 - DHCP: ☒ Static (DHCP request successful)
 - IPv4 Address: 192.168.1.1
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 0.0.0.0
 - DNS Server: 0.0.0.0
- IPv6 Configuration:
 - Automatic: ☒ Static
 - IPv6 Address: [Empty]
 - Link Local Address: FE80:210:11FF:FE0A:B032
 - Default Gateway: [Empty]
 - DNS Server: [Empty]
- 802.1X:
 - Use 802.1X Security: ☒
 - Authentication: MD5
 - Username: [Empty]
 - Password: [Empty]
- Fire: ☒ Top

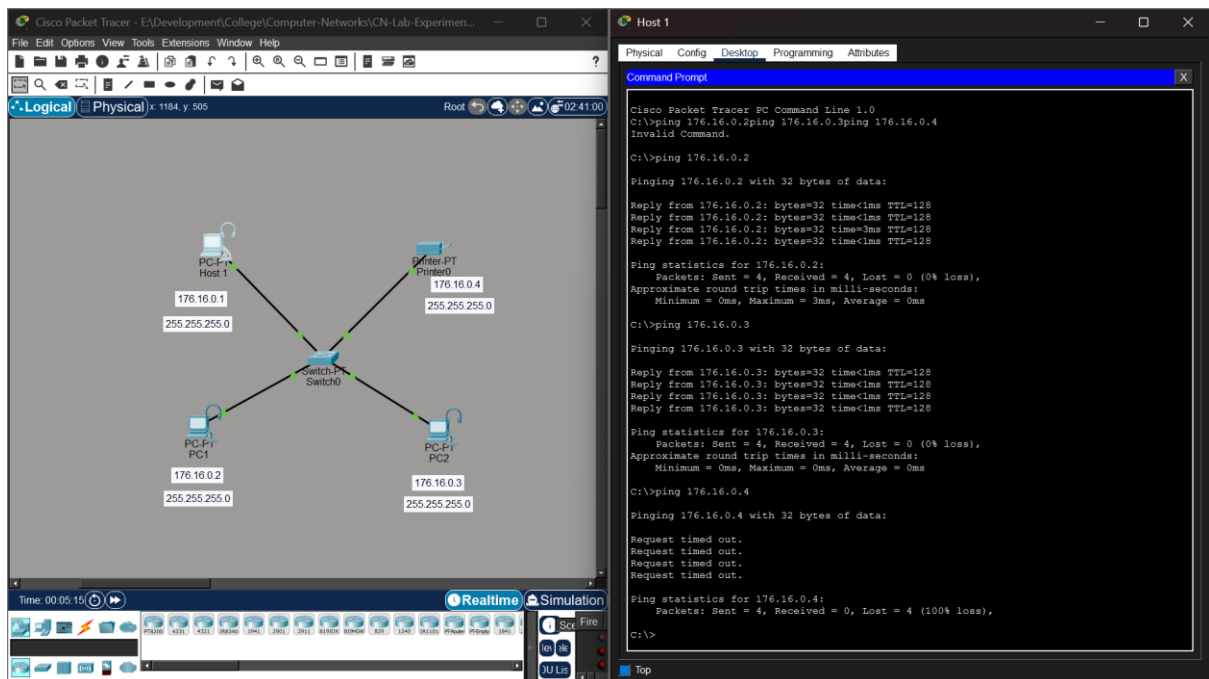
The image displays the Cisco Packet Tracer interface in the 'Physical' view. The network topology is the same as in the first image, showing a central '2960 24-T Switch' connected to three 'PC-PT' devices (PC0, PC1, PC2) and a 'Server-PT Server0'. The time is 00:08:46. The 'Realtime' button is highlighted in the bottom right corner. The 'Fire' section is set to 'Last Status'.

Physical View:

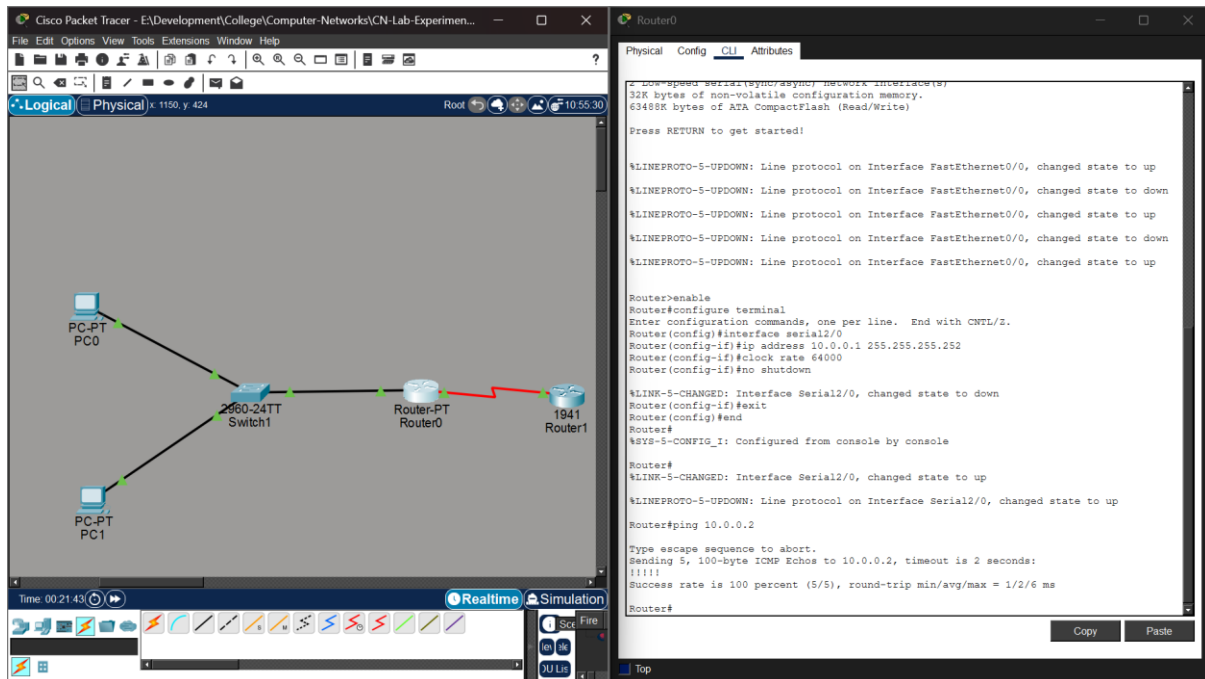
- Time: 00:08:46
- Realtime: ☒ Simulation
- Fire: ☒ Last Status



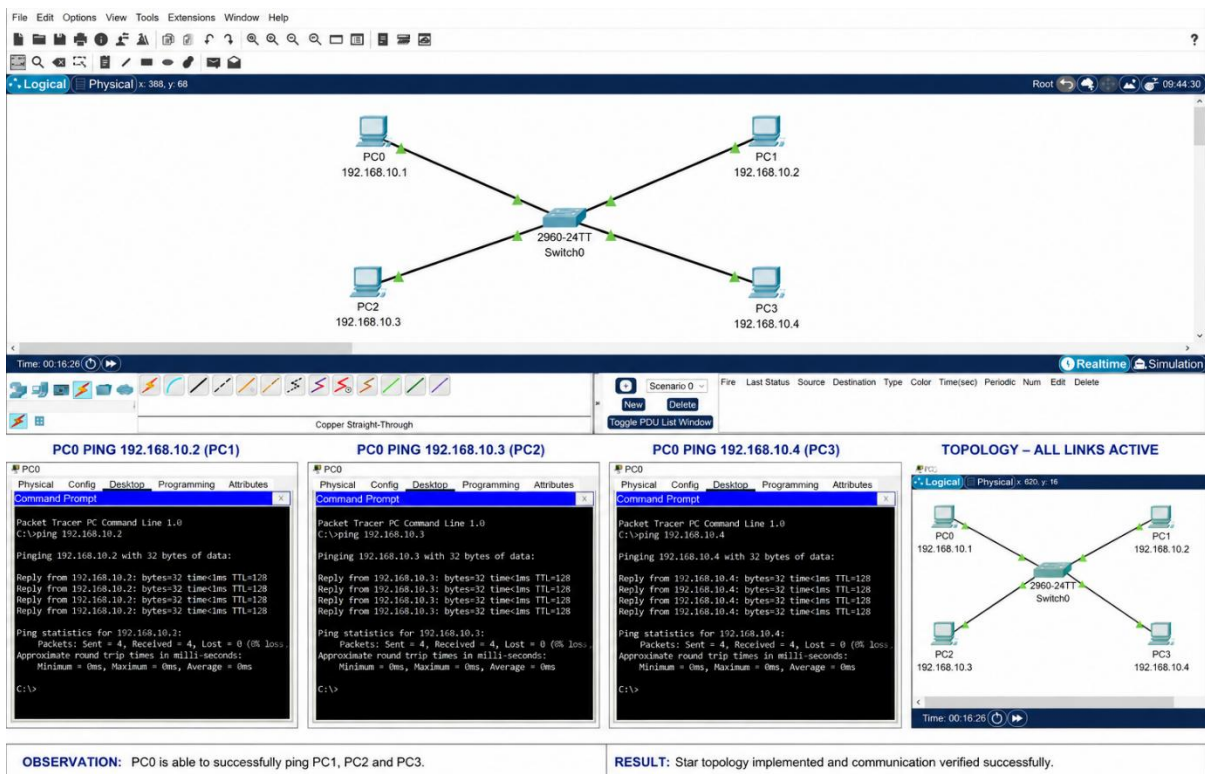
Experiment-16- Fire Wall



Experiment 17 – Firewall Configuration and Network Connectivity



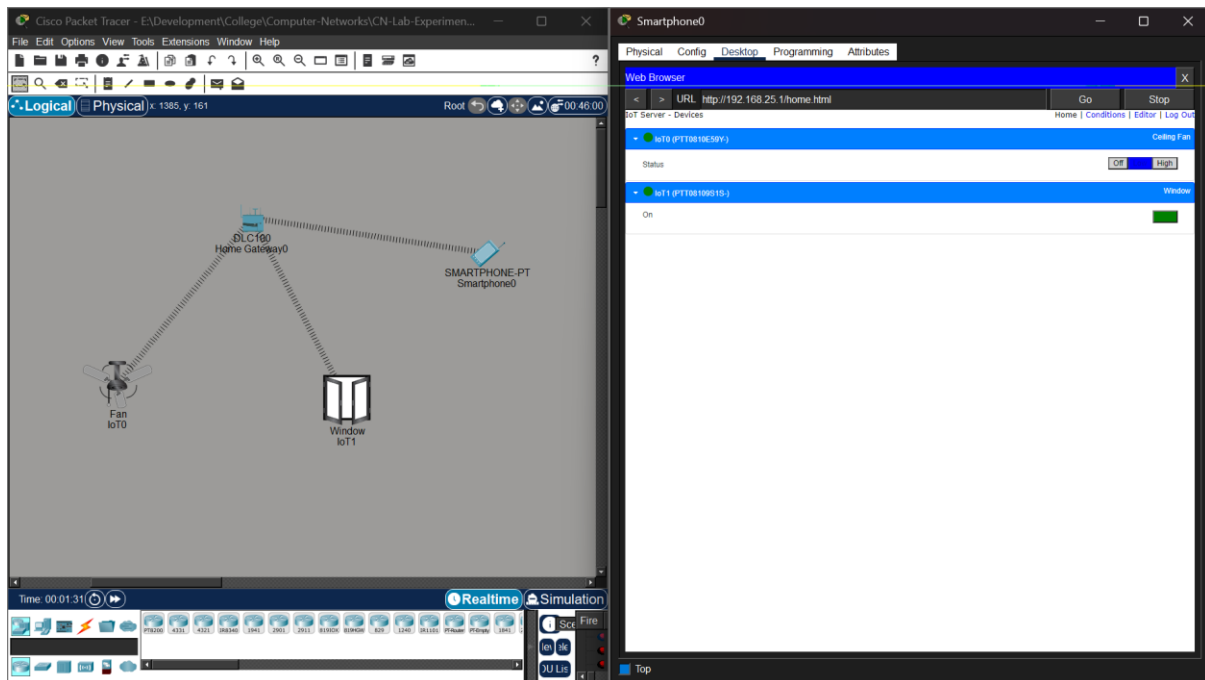
Experiment-18-Multimedia Network



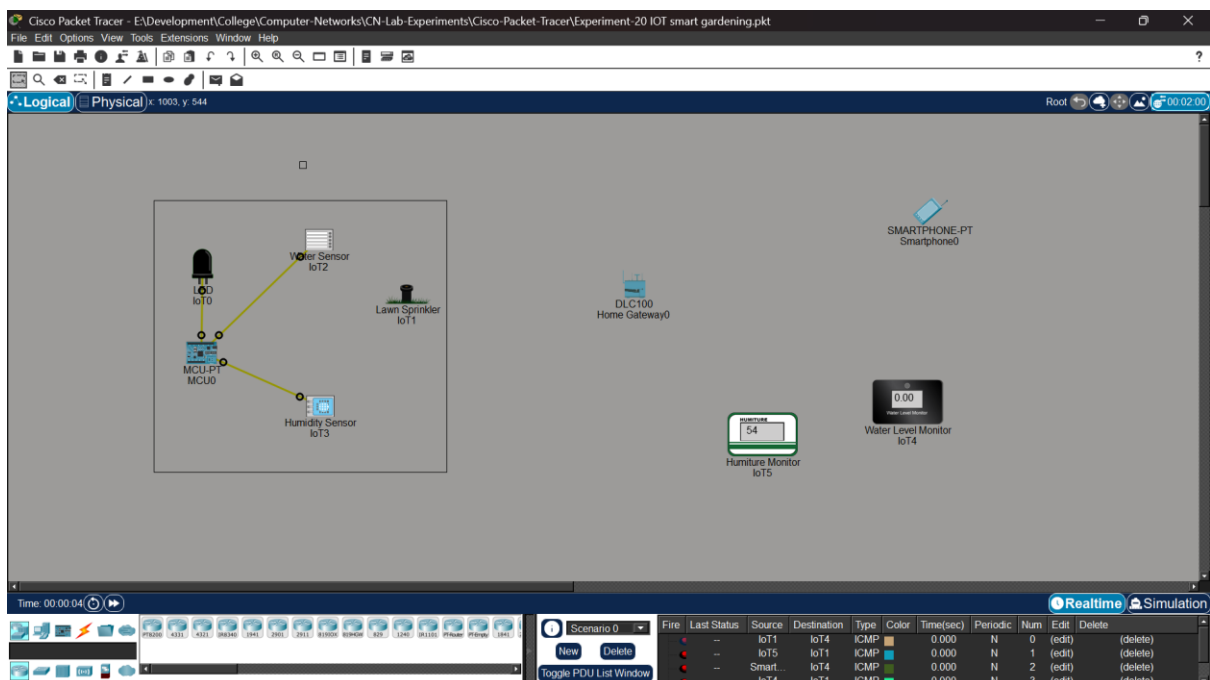
OBSERVATION: PC0 is able to successfully ping PC1, PC2 and PC3.

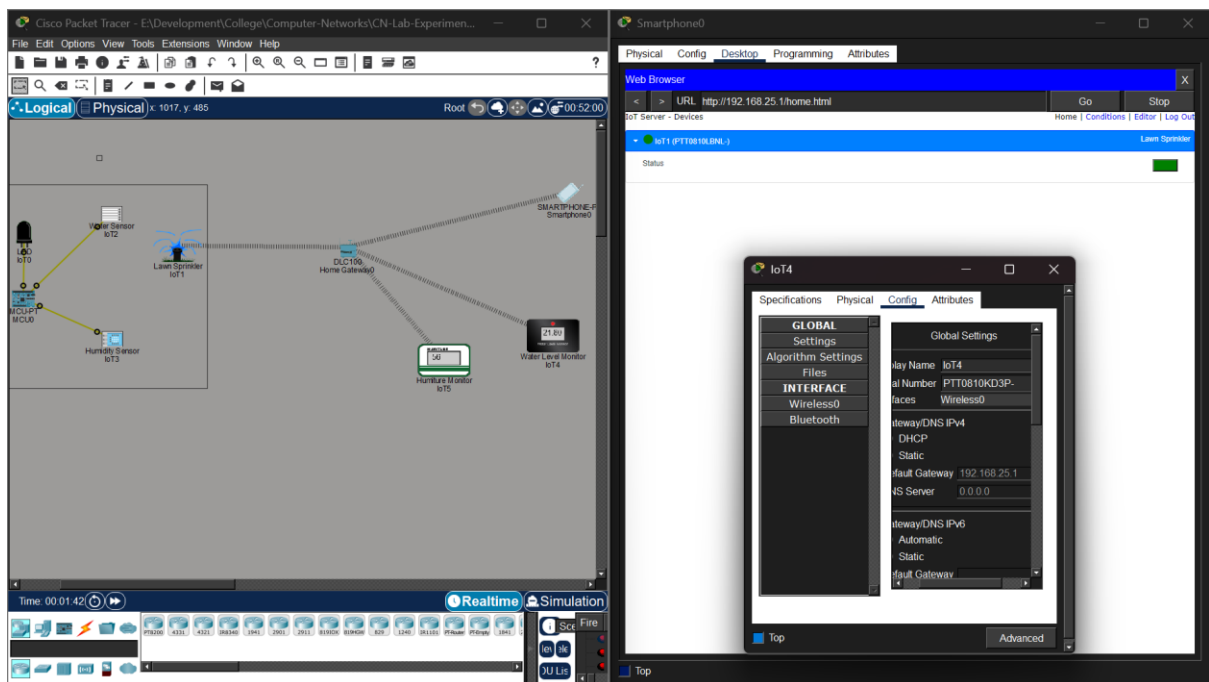
RESULT: Star topology implemented and communication verified successfully.

Experiment-19- IoT based smart home applications

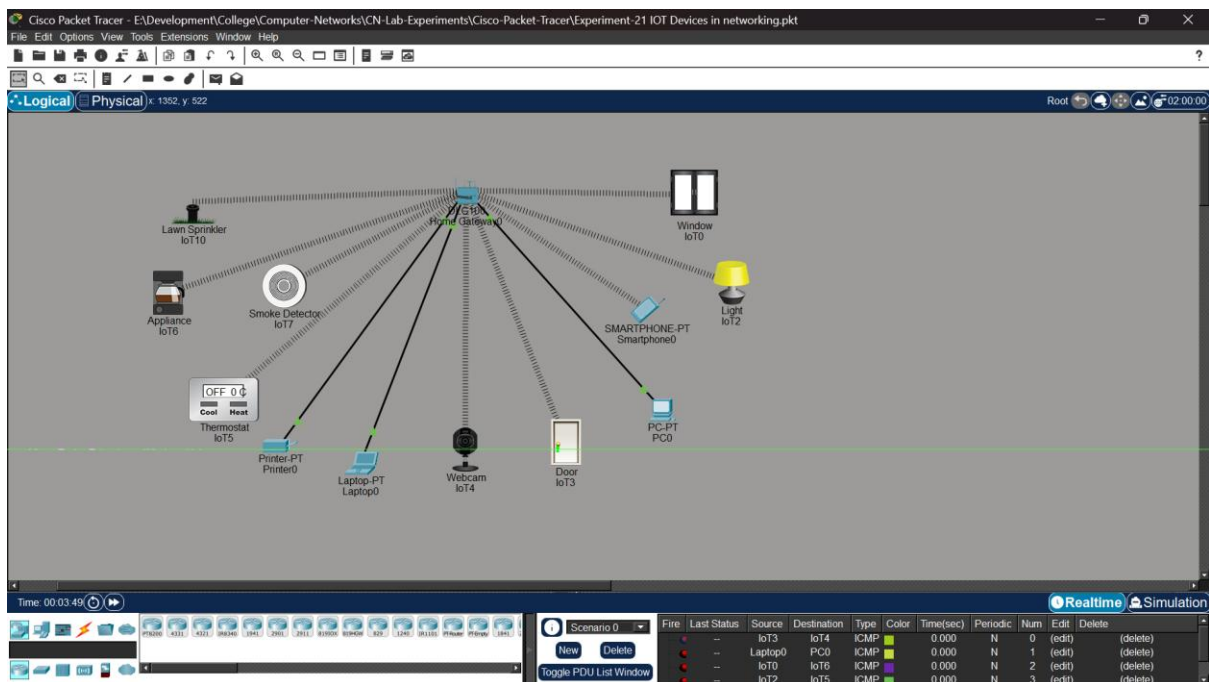


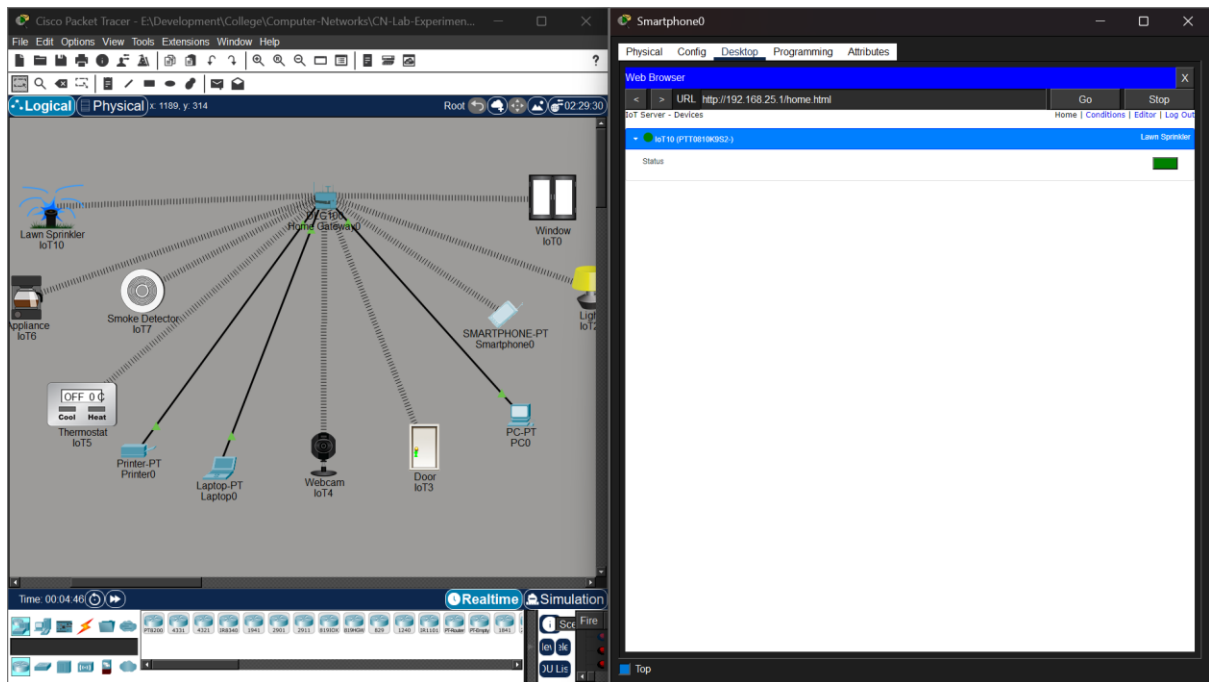
Experiment-20- IoT based smart home applications





Experiment-21- IoT devices in networking





Experiment-22-IoT based AAA Local and Server

Cisco Packet Tracer - E:\Development\College\Computer-Networks\CN-Lab-Experiments\Cisco-Packet-Tracer\Experiment-22-IoT based AAA Local and Server.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x 393, y 333

Root 03:10:30

Time: 00:49:58

Realtime Simulation

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	Router0	PC0	ICMP		0.000	N	0	(edit)	(delete)	
Successful	Router0	Server0	ICMP		0.000	N	1	(edit)	(delete)	
Successful	PC0	Server0	ICMP		0.000	N	2	(edit)	(delete)	

Scenario 0

New Delete

Toggle PDU List Window

Cisco Packet Tracer - E:\Development\College\Computer-Networks\CN-Lab-Experiments\Cisco-Packet-Tracer\Experiment-22-IoT based AAA Local and Server.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x 1503, y 543

Root 03:02:30

Time: 00:49:49.853

PLAY CONTROLS

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.002	Router0
	0.002	Switch0
	0.002	Switch0
	0.002	Switch0
	0.002	-
	0.003	Switch0
	0.003	PC0
	0.003	-
	0.004	Switch0
	0.004	Server0
	0.004	Switch0
	0.005	Server0
	0.006	Switch0
	0.006	Switch0

Reset Simulation

Constant Delay

Captured to: 0.000 s

Play Controls

Event List Filters: Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.223, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, ICG, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profile, RADIUS, RDP, RFP, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters

Show All/None

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	Router0	PC0	ICMP		0.000	N	0	(edit)	(delete)	
Successful	Router0	Server0	ICMP		0.000	N	1	(edit)	(delete)	
Successful	PC0	Server0	ICMP		0.000	N	2	(edit)	(delete)	